

Work distribution and fluctuation theorem in AdS/CFT

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Abstract

In quantum mechanics, the Crooks fluctuation theorem (FT) concerning the work distribution $p(W)$ has been proven by Tasaki. We propose a method to compute $p(W)$ on the AdS gravity through AdS/CFT and its FT is confirmed for a gravity model. We also discuss the bulk meaning of $p(W)$ and FT.

1. Thermodynamics in AdS/CFT

In the context of AdS/CFT, ...

	CFT	AdS	
State	$\rho_{\text{th}} = e^{-\beta H}/Z$	Static BH	[Gubser, Klebanov, Peet ('96)], [Witten ('98)], [Ryu, Takayanagi ('06)], ...
Entropy	$-\text{Tr} \rho_{\text{th}} \log \rho_{\text{th}}$	$\frac{\text{Area}}{4G}$	
2nd law	$\dot{S}_{\text{c.g.}}(t) \geq 0$	$\delta(\text{Area}) \geq 0$	[Engelhardt, Wall ('19)], [Chandrasekaran, Penington, Witten ('23)], [Takeda ('24)]
FT	Tasaki-Crooks	???	This work

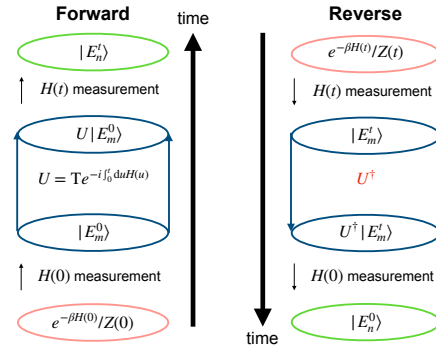
Another approach to gravitational FT not via AdS/CFT: [Iso, Okazawa ('08)]

FT explains 2nd law mesoscopically → clue to quantum gravity?

2. Tasaki-Crooks FT

Protocol: 2pt measurement

[Tasaki ('00)], [Kurchan ('00)]



Work distribution

$$\text{Forward transition probability } p_{m \rightarrow n} = |\langle E_n^i | U | E_m^f \rangle|^2 \frac{e^{-\beta E_m^f}}{Z(0)}$$

$$\text{Forward work distribution } p_F(W) := \sum_{m,n} p_{m \rightarrow n} \delta(W - E_n^i + E_m^f)$$

For reverse process, $p_R(W)$ is similarly defined

The fluctuation theorem

$$p_R(-W) = e^{-\beta(W - \Delta F)} p_F(W) \quad e^{\beta \Delta F} := \frac{Z(0)}{Z(t)}$$

$$\Rightarrow \text{Integral form } 1 = \langle e^{-\beta(W - \Delta F)} \rangle \Rightarrow \text{2nd law } \langle W \rangle \geq \Delta F$$

Characteristic function

[Talkner, Hänggi (2007)]

$$G_F(u) := \int dW e^{iuW} p_F(W) = \langle e^{iuW} \rangle_F, \quad G_R(u) := \langle e^{iuW} \rangle_R$$

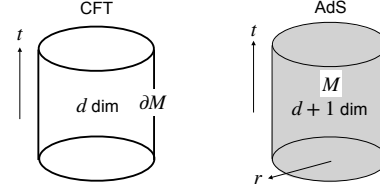
$$\text{Equivalently, } G_F(u) = \text{Tr} \left[U^\dagger e^{iuH(t)} U e^{-iuH(0)} \frac{e^{-\beta H(0)}}{Z(0)} \right]$$

→ Path integral on a Schwinger-Keldysh contour

3. Bulk expression

GKP-W dictionary

$$\int \mathcal{D}\phi e^{iS_{\text{CFT}}[\partial M; \phi] + i \int d^d x J(x) O_\Delta(x)} = e^{iS_{\text{AdS}}[M; \Phi_{\text{cl}}]} \Big|_{\Phi \sim r^{\Delta-d} J}$$



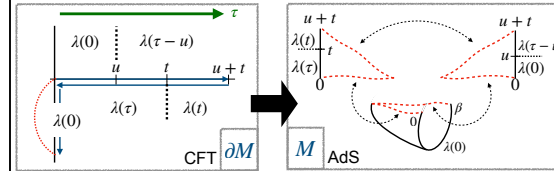
M can be generic Schwinger-Keldysh contours: [Skenderis, van Rees ('08)]

$$\text{Below, we consider } H(u) = H_{\text{CFT}} + \int d\vec{x} \lambda(u, \vec{x}) O(\vec{x})$$

Bulk expression of $G(u)$ [Takeda ('25)]

$$G_F(u) := \langle e^{iuW} \rangle_F = \text{GKP-W dictionary}$$

$$\int \mathcal{D}\phi e^{iS_{\text{CFT}}[\partial M; \phi] + i \int d^d x \lambda(x) O(x)} = e^{iS_{\text{AdS}}[M; \Phi_{\text{cl}}]} \Big|_{\Phi \sim r^{\Delta-d} \lambda}$$



✓ Impose "smooth" boundary condition on the glueing surfaces.

✓ Get $p_F(W)$ by inverse Fourier transform. [Skenderis, van Rees ('08)]

✓ Repeat a similar computation for $p_R(W)$.

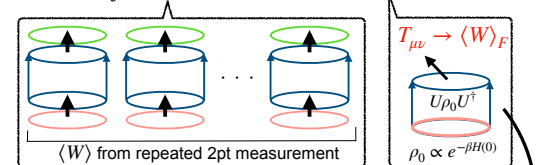
✓ If AdS/CFT holds, the bulk $p_{FIR}(W)$ must satisfy FT as well.

FT was checked for free scalar on BTZ black hole perturbatively in λ

4. Meaning for bulk realtime physics

Let us focus on $\langle W \rangle_F$. By simple algebra, we have in the CFT side,

$$\langle W \rangle_F = \int dW W p_F(W) = \text{Tr} [H(t) U \rho_0 U^\dagger] - \text{Tr} [H(0) \rho_0]$$



✓ $\langle W \rangle_F$ can be computed in two ways in bulk.

• From the above M

• From the ordinary initial value problem

✓ Both coincide for the above BTZ example.

➡ Physics on M knows some information of the realtime evolution

➡ Does $p_F(W)$ know "unraveling" of this realtime evolution?

Ordinary AdS/CFT

$$\delta g_{\mu\nu} \rightarrow \langle W \rangle_F$$

Initial value problem

BH

λ